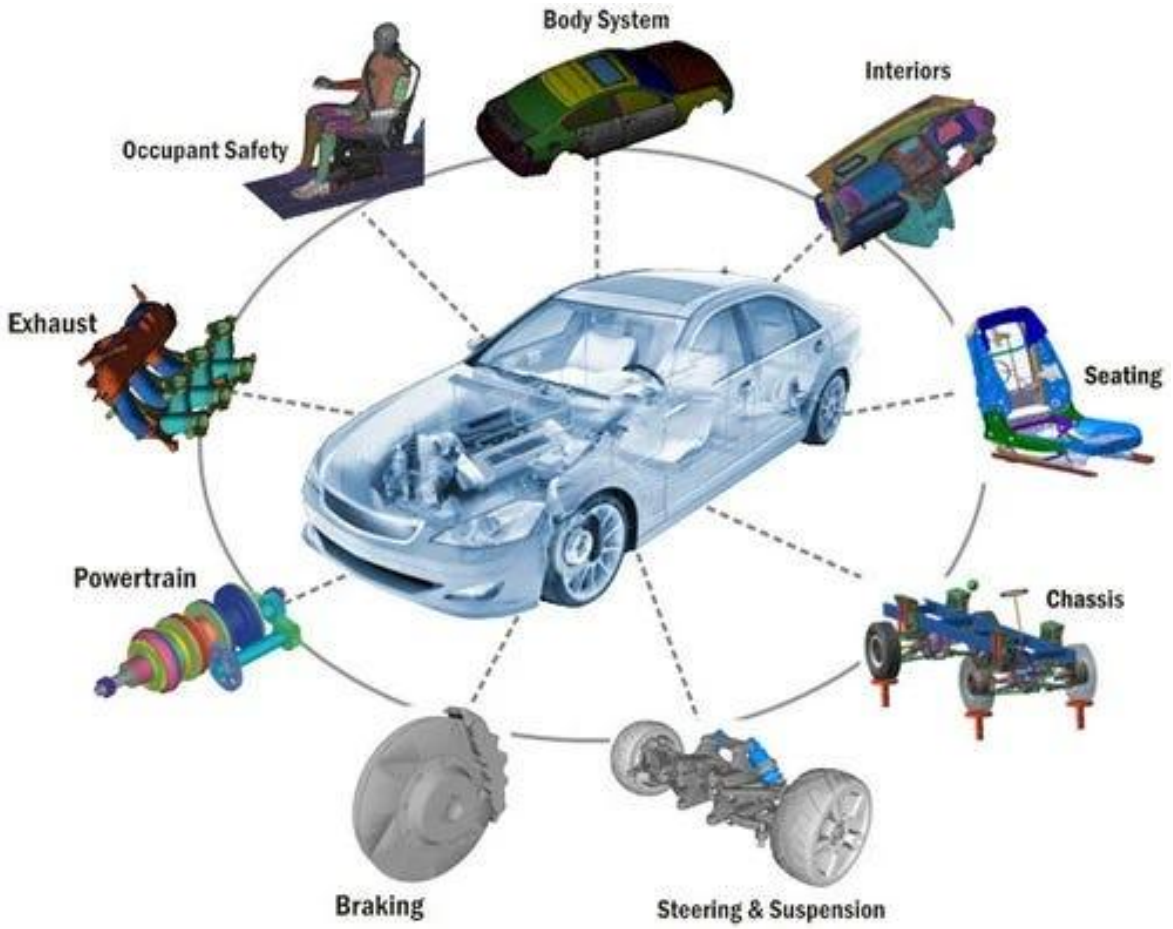
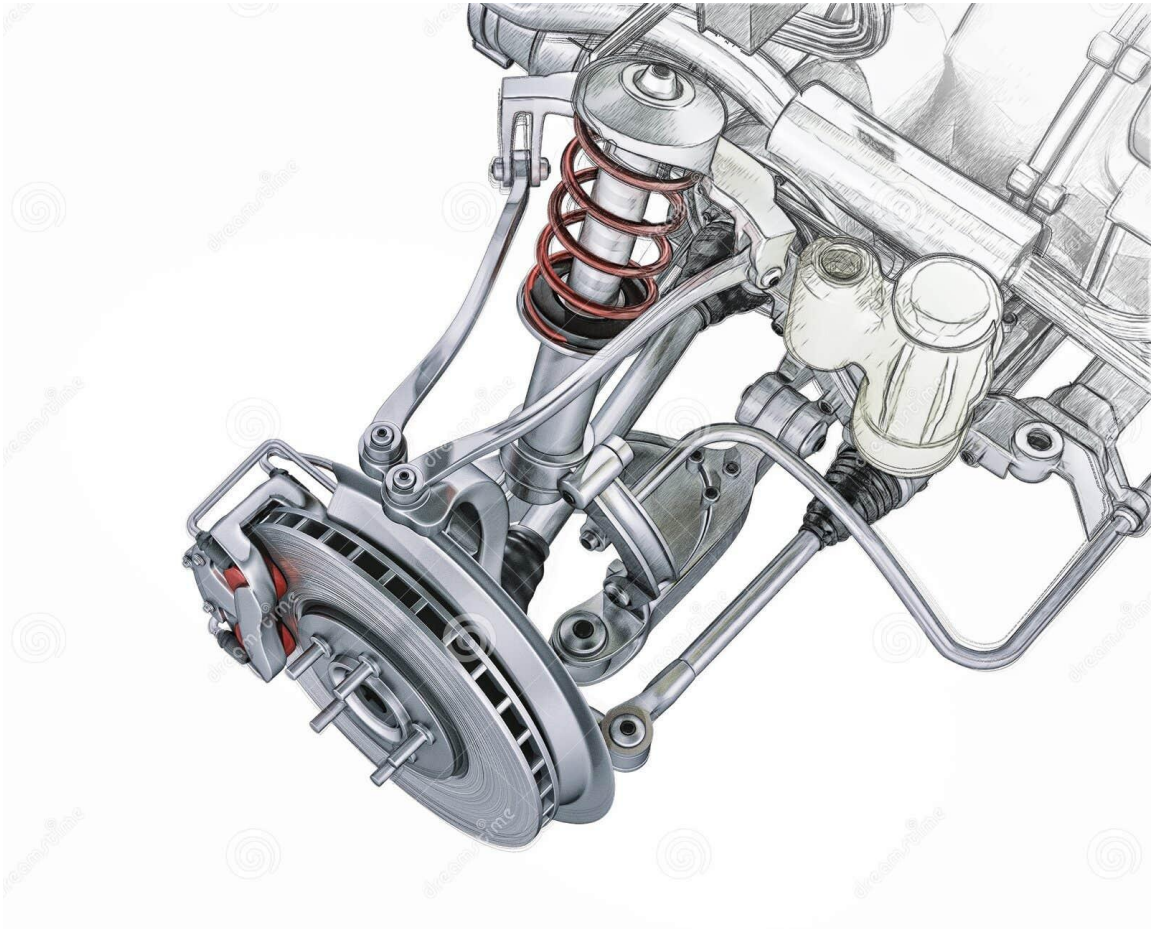




Automotive Engineering (MAM 336)

Week 1
Introduction to Automotive Engineering

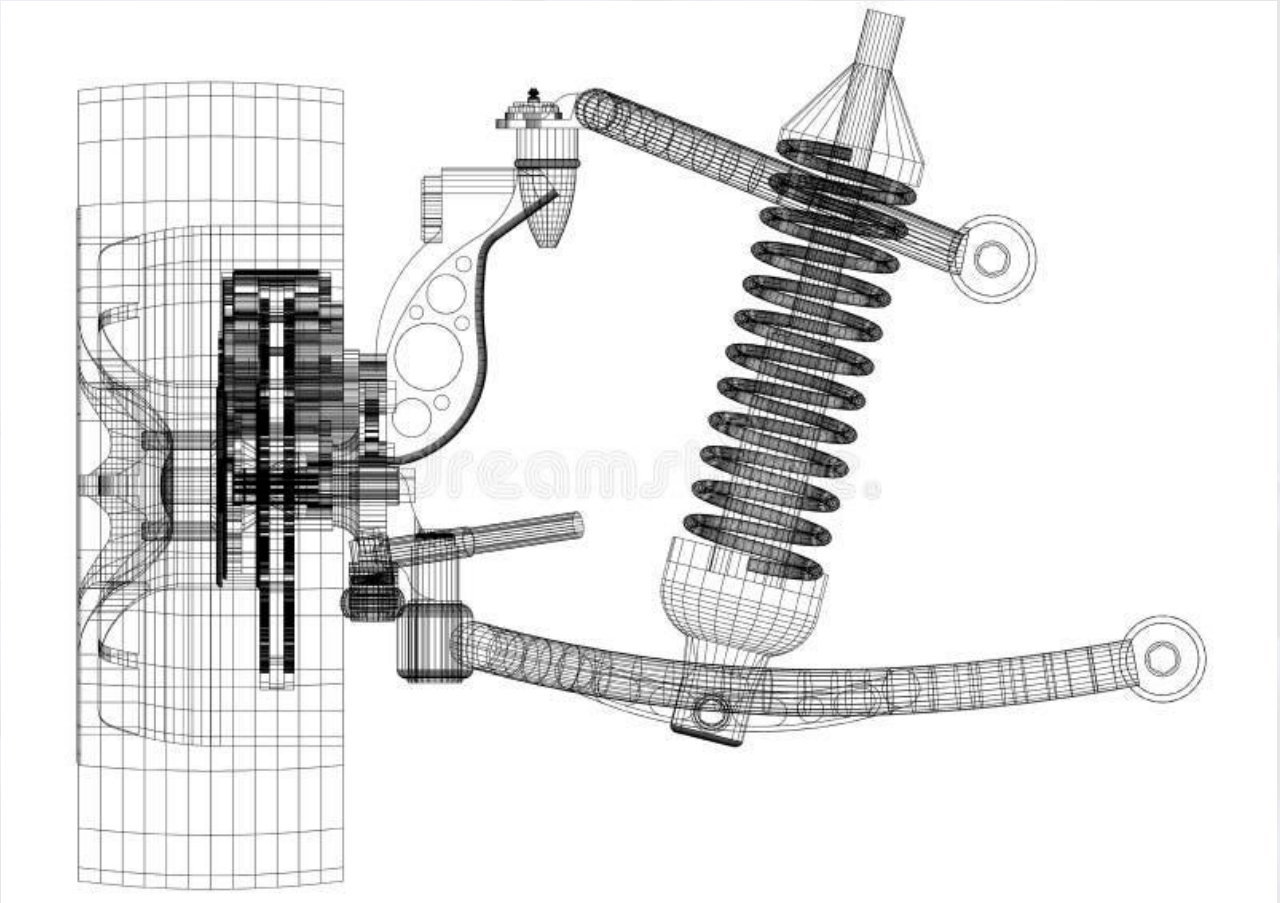
DR/ MOHAMED EL TAHER MOHAMED

Lecture Outline

1. Welcome and Course Outline

2. Course basics

3. Introduction to **Automotive Engineering**



1. Welcome

Code	Course Title	Pre-req	Cr. Hrs.	Ct. Hr.				Assessment			
				Lec.	Lab	Tut	Sum	MT1	MT2	SA	Final
MAM 336	Automotive Engineering	MAM 301	3	2	0	2	4	30	20	10	40
Course Content	Characteristics of Ground Vehicle, Classification of Motor Vehicle. Manual and automatic Transmission Systems, Propeller Shaft and Drive Shaft. Tires, Construction of Tire, Tire Dynamics. Types of Suspension System: Mechanical, Pneumatic and Hydraulic suspension systems. Design Analysis of Suspension System, Braking System, Steering System, introduction of hybrid cars, autonomous cars.										
References	• Abubakar, S.; Alammari, Youssef; Kaisan, M. U.; Mahroogi, Faisal O.; Narayan, S.; Sakthivel, R, 2019, " An introduction to automotive engineering", John Wiley & Sons & Scrivener Publishing.										

1. Welcome

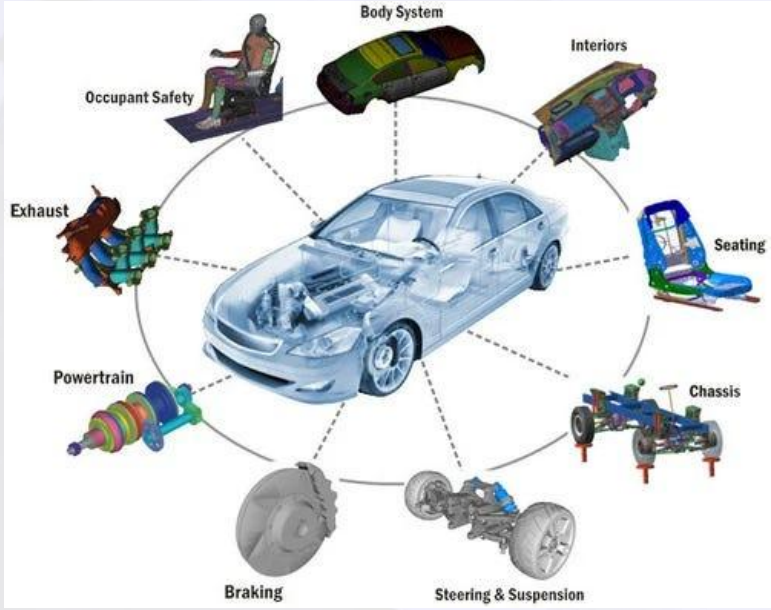
Identify major classes of motor vehicles and their principal applications.

Explain the role of the power-train and the major non-power-train systems.

Describe how transmission, tires, suspension, braking and steering systems support vehicle motion.

Recognize the basic engineering concepts behind hybrid and autonomous vehicles.

Connect vehicle performance, handling, ride and safety to system-level design.



What Is Automotive Engineering?

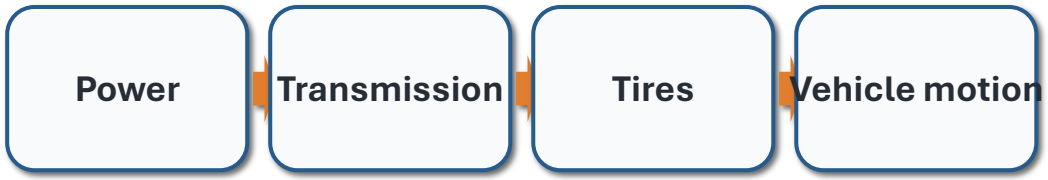
Automotive engineering applies engineering principles to the design, analysis, manufacture and operation of road vehicles.

Motor vehicles are used to transport goods or passengers or to perform specific functions on land.

The course focuses on non-guided ground vehicles and their major mechanical and vehicle systems.

Automotive engineering integrates mechanics, materials, power transmission, fluids, controls and safety.

Vehicle = integrated system



Vehicle Classification

By guidance / running gear

Guided vehicles: move along a fixed guideway, e.g., railway vehicles.

Non-guided vehicles: can move in any direction; these are the main subject of the book.

Single-track: two wheels, with or without a sidecar.

Multi-track: three or more wheels.

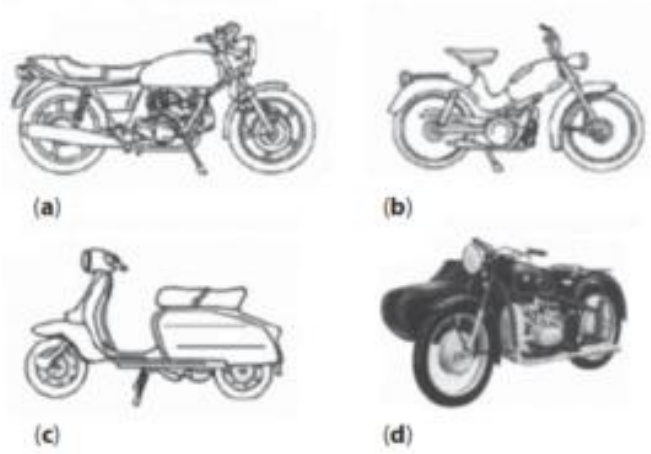


Figure 1.1 Single track motor vehicles.

By application

Passenger cars and buses.

Pick-ups, vans and trucks.

Tractors and special-purpose vehicles.

Transport vehicles can also be classified by body type and load capacity.

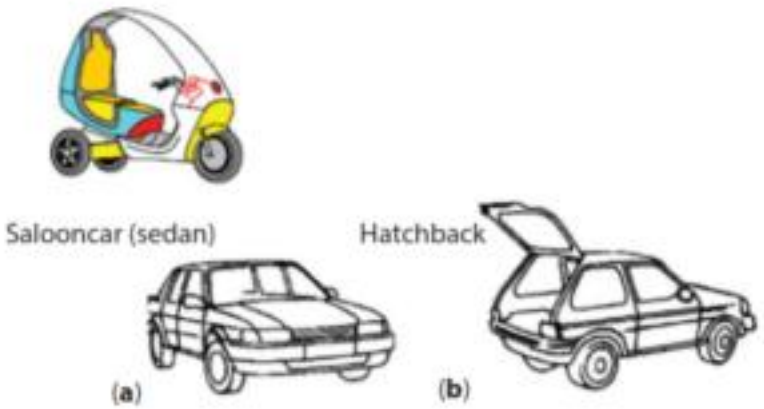


Figure 1.2 Multi-track motors.

Vehicle Types and Load Capacity

Passenger car: intended for carrying up to nine passengers with luggage.

Bus: designed to transport more than nine passengers and luggage.

City buses emphasize rapid passenger turnover through low steps, wide doors and low floors.

Tour buses are designed for long-distance comfort and provide substantial luggage space.

Truck classes in the source range from extra-light-duty vehicles to extra-heavy-duty vehicles.

Extra-light	**≤0.75 t**
Light	**0.75–2.5 t**
Medium	**2.5–5 t**
Heavy	**5–10 t**
Extra-heavy	**>10 t**

ISO-Based Vehicle Classification & Drive Layouts

ISO 3833 groups ground vehicles into categories such as motorbikes, traveler cars, buses, pickup trucks, agricultural machinery, trailer cars and semi-trailers.

ISO3833 classifies ground vehicles in 7 groups:

1. Motorbikes
2. Cars for travelers
3. Pick-up Buses
4. Pickup Trucks
5. Agricultural machinery
6. Trailer cars
7. Semi-trailers

Drive arrangements described in the source include front-engine/front-wheel drive, front-engine/rear-wheel drive, rear-engine/rear-wheel drive and four-wheel drive.

Engine/transmission placement strongly influences packaging and power flow.

ISO-Based Vehicle Classification & Drive Layouts

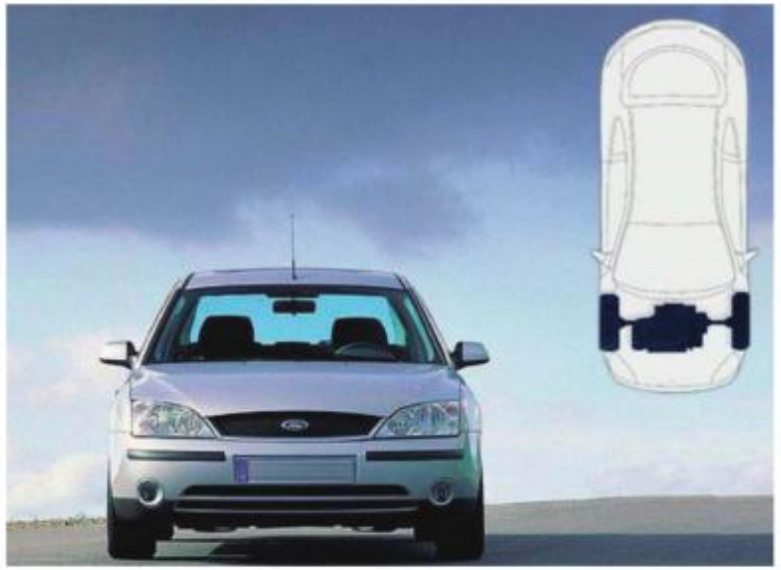


Figure 1.1 Front-engine, front-wheel drive

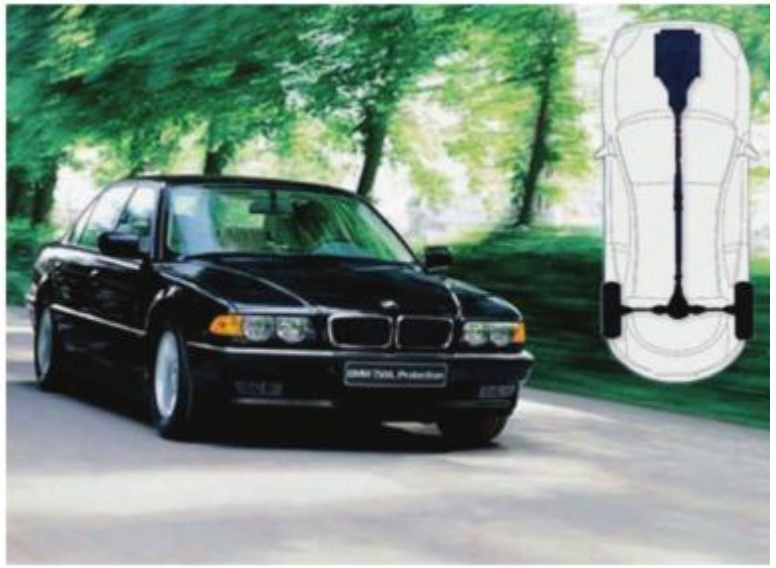


Figure 1.2 Front-engine, rear-wheel drive

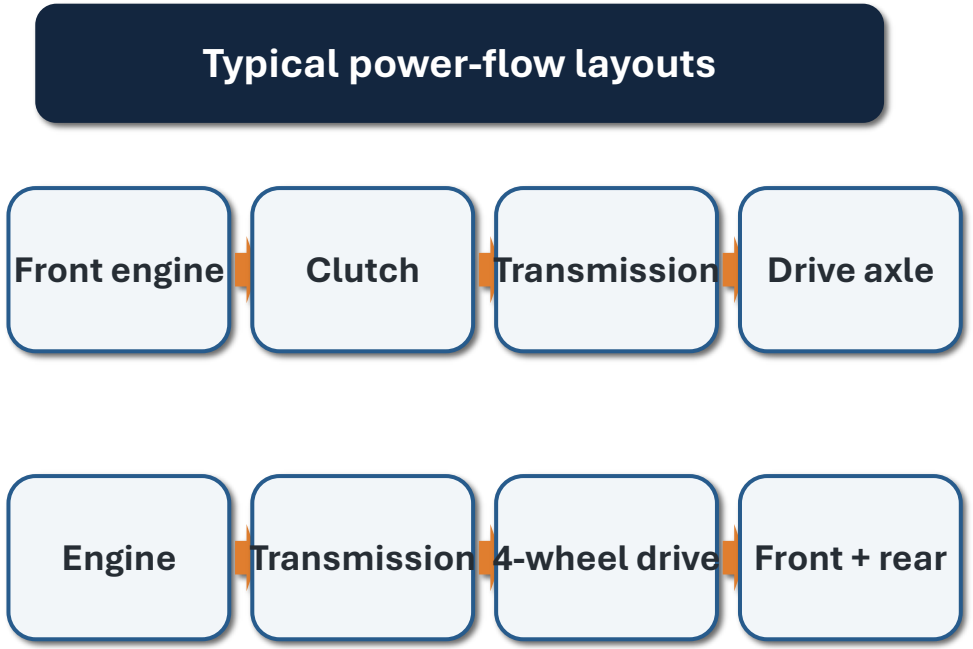


Figure 1.3 Rear-engine, rear-wheel drive



Figure 1.4 Mid-engine, rear-wheel drive

ISO-Based Vehicle Classification & Drive Layouts



Major Automobile Systems

Power plant / engine

Drive train: clutch, gearbox, shafts and differential

Electrical system: ignition, starting, lighting and battery

Wheels and tires

Suspension system

Steering system

Braking system

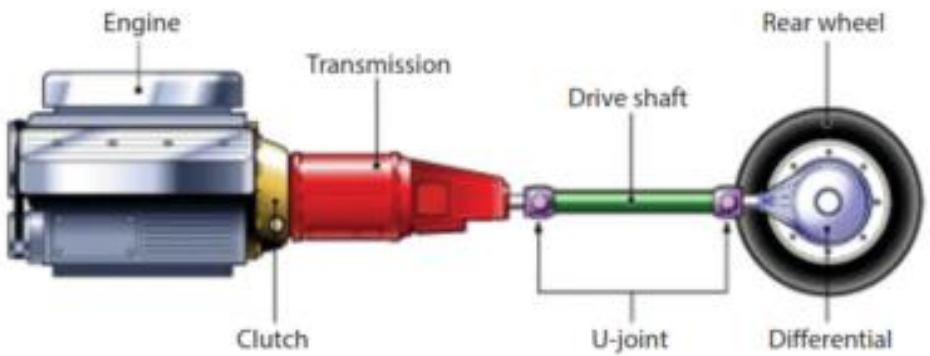
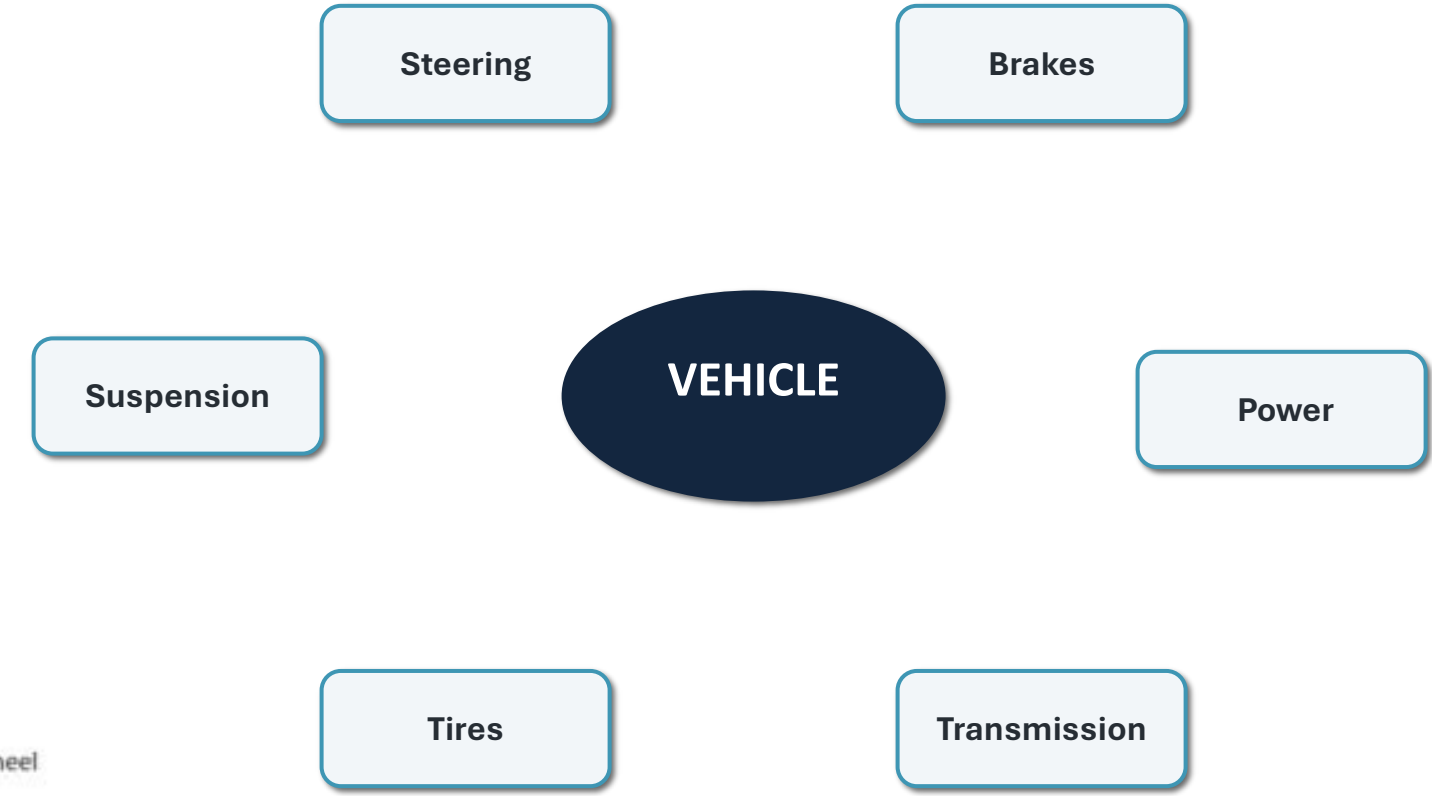
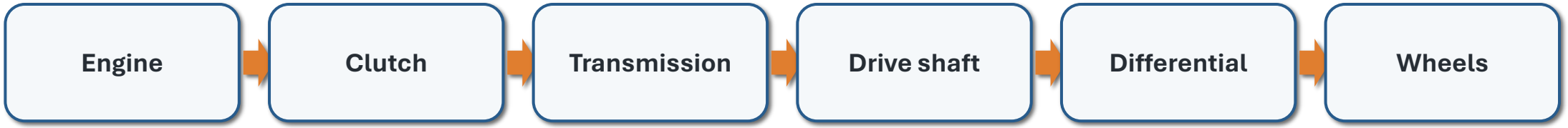


Figure 1.7 Transmission system.



Power Flow Through the Vehicle

The drive train indicates the power flow from the engine to the driving wheels.



Power source

Torque / speed conversion

Torque distribution

Clutch: Function and Need

Engages and disengages the engine from the drive.

Allows the driver to control power flow between the engine and transmission/transaxle.

Works primarily through friction between rotating members.

Allows gradual engagement of rotating parts.

Provides positive linkage capable of transmitting engine torque.

Separates the engine from the drive train and re-engages it when required.

Basic clutch arrangement



Friction → torque transmission



Thank You!